

AKADEMIA GÓRNICZO-HUTNICZA
IM. STANISŁAWA STASZICA W KRAKOWIE

AGH UNIVERSITY OF SCIENCE
AND TECHNOLOGY

Contact Structural Analysis in Marc Mentat

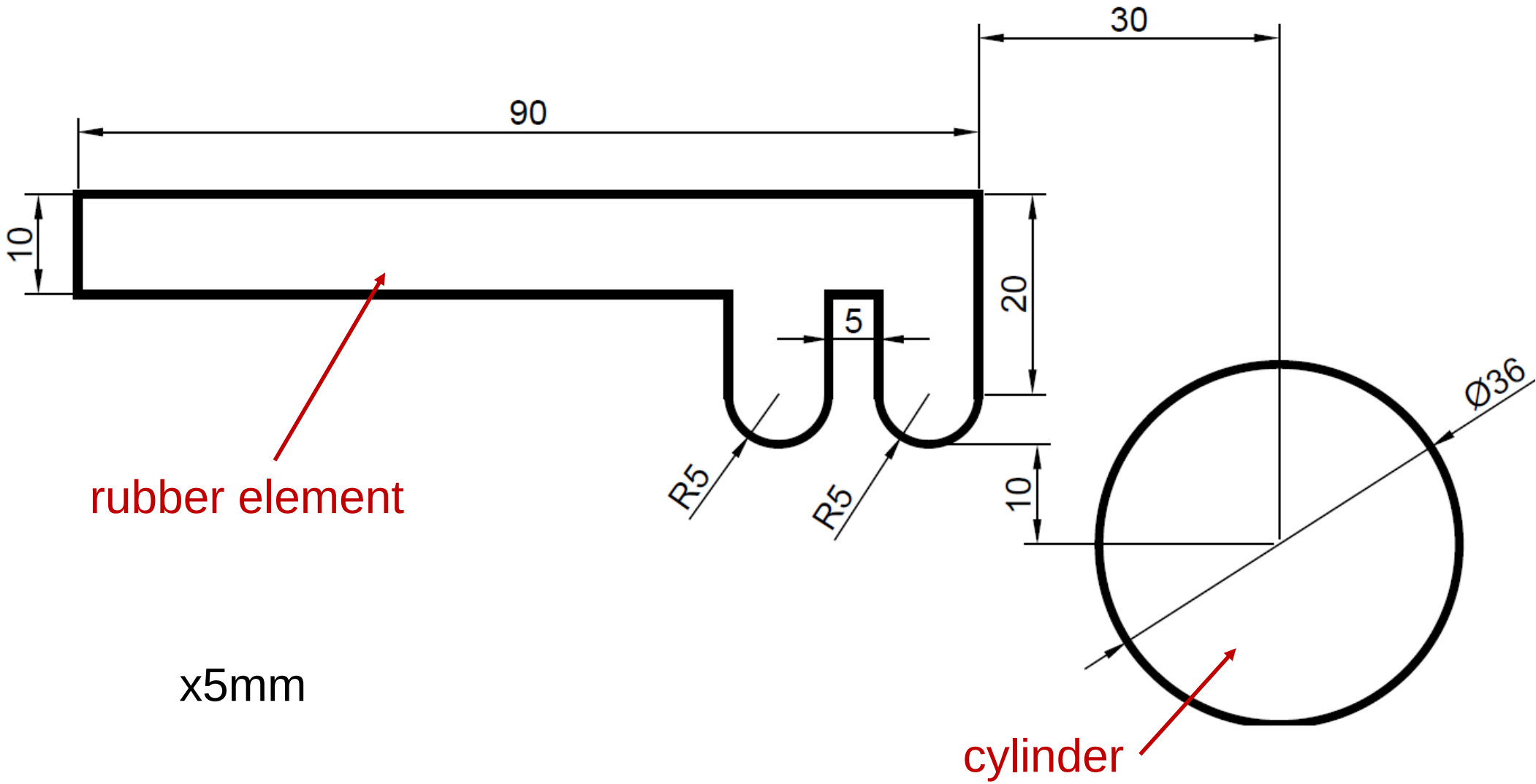
Created by Jakub Bryła

Agenda:

1. Geometry
2. Material Properties
3. Contact
4. Boundary condition
5. Loadcases
6. Job
7. Run job and save results
8. Modification in Contact
9. Run job and save results

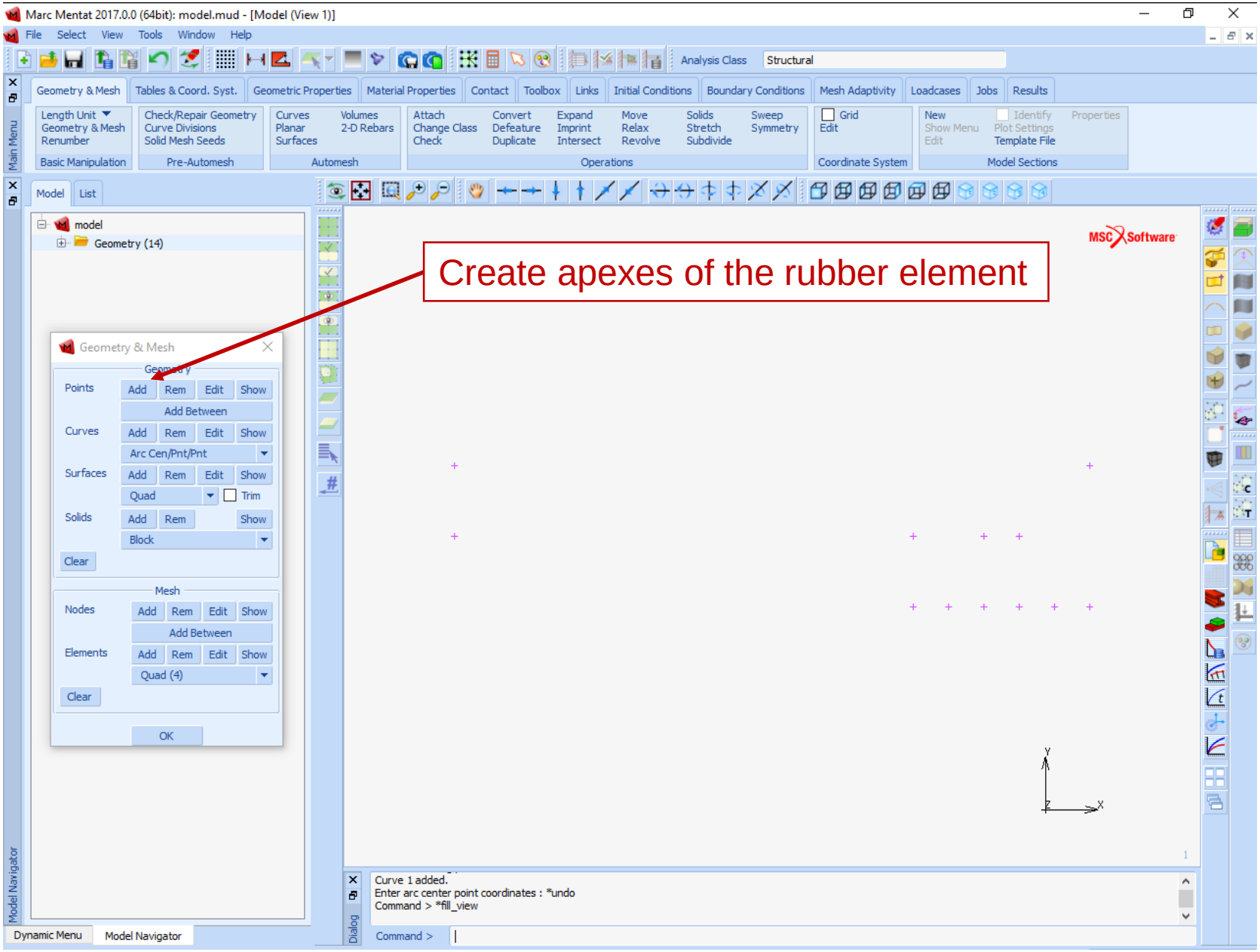
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Geometry



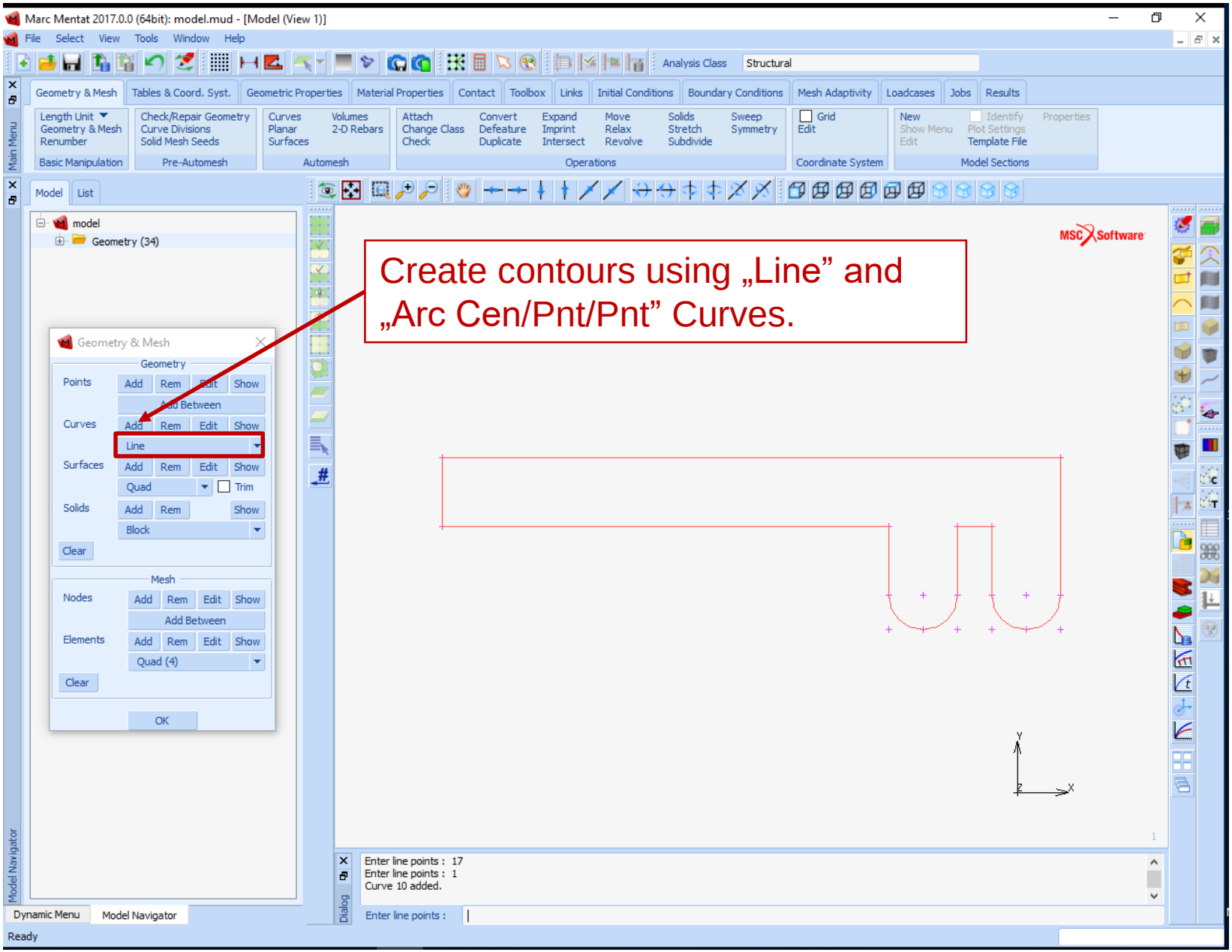
x5mm

Geometry

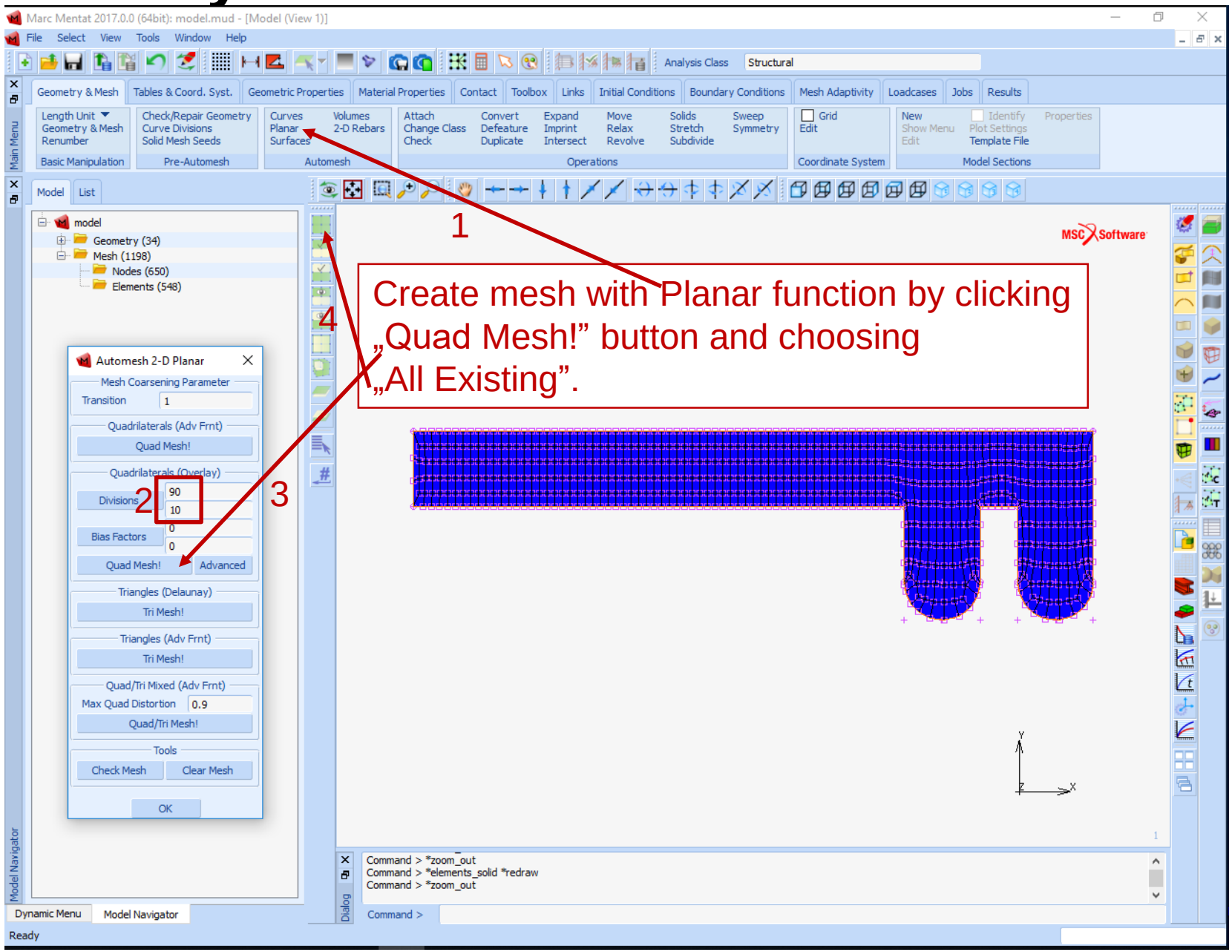


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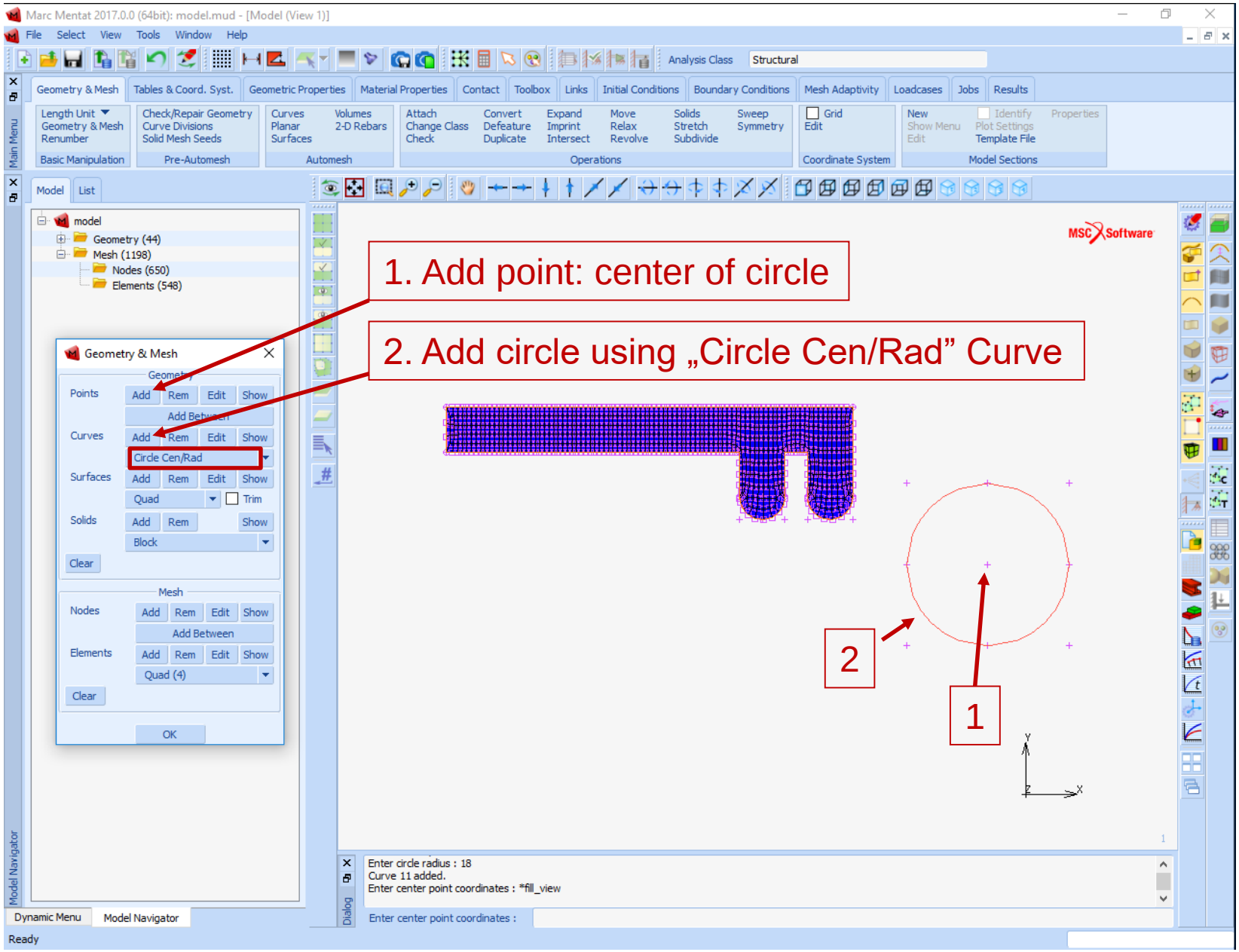
Geometry


Marc Mentat 2017.0.0 (64bit): model.mud - [Model (View 1)]
File Select View Tools Window Help
Analysis Class Structural
Geometry & Mesh
Tables & Coord. Syst.
Geometric Properties
Material Properties
Contact
Toolbox
Links
Initial Conditions
Boundary Conditions
Mesh Adaptivity
Loadcases
Jobs
Results
Model
List
model
Geometry (34)
Geometry & Mesh
Geometry
Points Add Rem Edit Show
Add Between
Curves Add Rem Edit Show
Line
Surfaces Add Rem Edit Show
Quad Trim
Solids Add Rem Show
Block
Clear
Mesh
Nodes Add Rem Edit Show
Add Between
Elements Add Rem Edit Show
Quad (4)
Clear
OK
Create contours using „Line” and „Arc Cen/Pnt/Pnt” Curves.
MSC Software
Enter line points : 17
Enter line points : 1
Curve 10 added.
Enter line points :
Dynamic Menu
Model Navigator
Ready

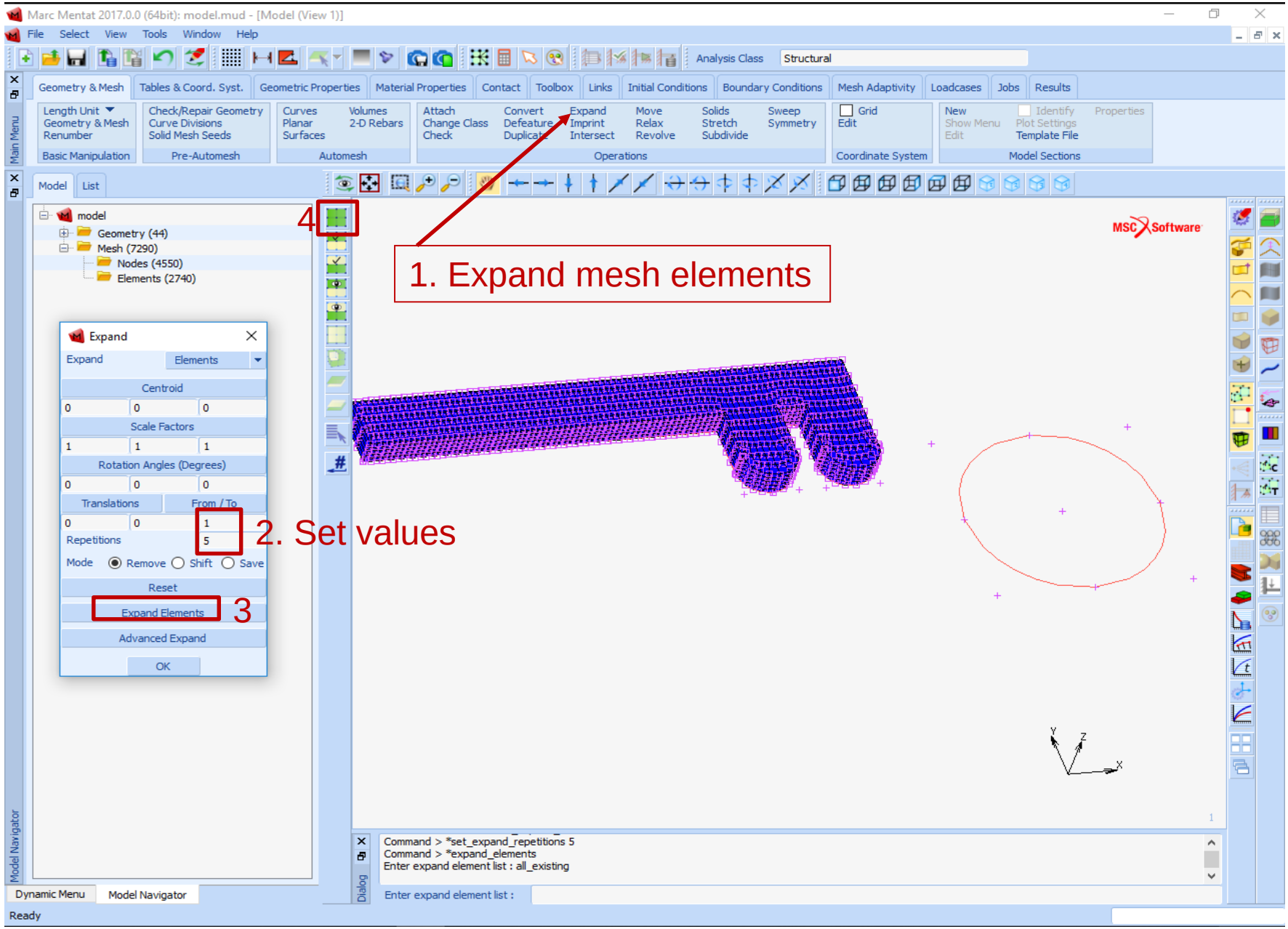
Geometry


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File Select View Tools Window Help
Analysis Class Structural
Geometry & Mesh | Tables & Coord. Syst. | Geometric Properties | Material Properties | Contact | Toolbox | Links | Initial Conditions | Boundary Conditions | Mesh Adaptivity | Loadcases | Jobs | Results
Length Unit
Geometry & Mesh
Renumber
Basic Manipulation
Check/Repair Geometry
Curve Divisions
Solid Mesh Seeds
Pre-Automesh
Curves
Planar
Surfaces
Volumes
2-D Rebars
Automesh
Attach
Change Class
Check
Convert
Defeature
Duplicate
Expand
Imprint
Intersect
Move
Relax
Revolve
Solids
Stretch
Subdivide
Sweep
Symmetry
Operations
Grid
Edit
Coordinate System
New
Show Menu
Edit
Identify
Plot Settings
Template File
Properties
Model Sections
Model
List
model
Geometry (34)
Mesh (1198)
Nodes (650)
Elements (548)
1
Create mesh with Planar function by clicking „Quad Mesh!” button and choosing „All Existing”.
4
Automesh 2-D Planar
Mesh Coarsening Parameter
Transition 1
Quadrilaterals (Adv Frnt)
Quad Mesh!
Quadrilaterals (Overlay)
Divisions 90 10
Bias Factors 0 0
Quad Mesh!
Advanced
2
3
Triangles (Delaunay)
Tri Mesh!
Triangles (Adv Frnt)
Tri Mesh!
Quad/Tri Mixed (Adv Frnt)
Max Quad Distortion 0.9
Quad/Tri Mesh!
Tools
Check Mesh
Clear Mesh
OK
MSC Software
Command > *zoom_out
Command > *elements_solid *redraw
Command > *zoom_out
Command >
Dynamic Menu
Model Navigator
Ready

Geometry

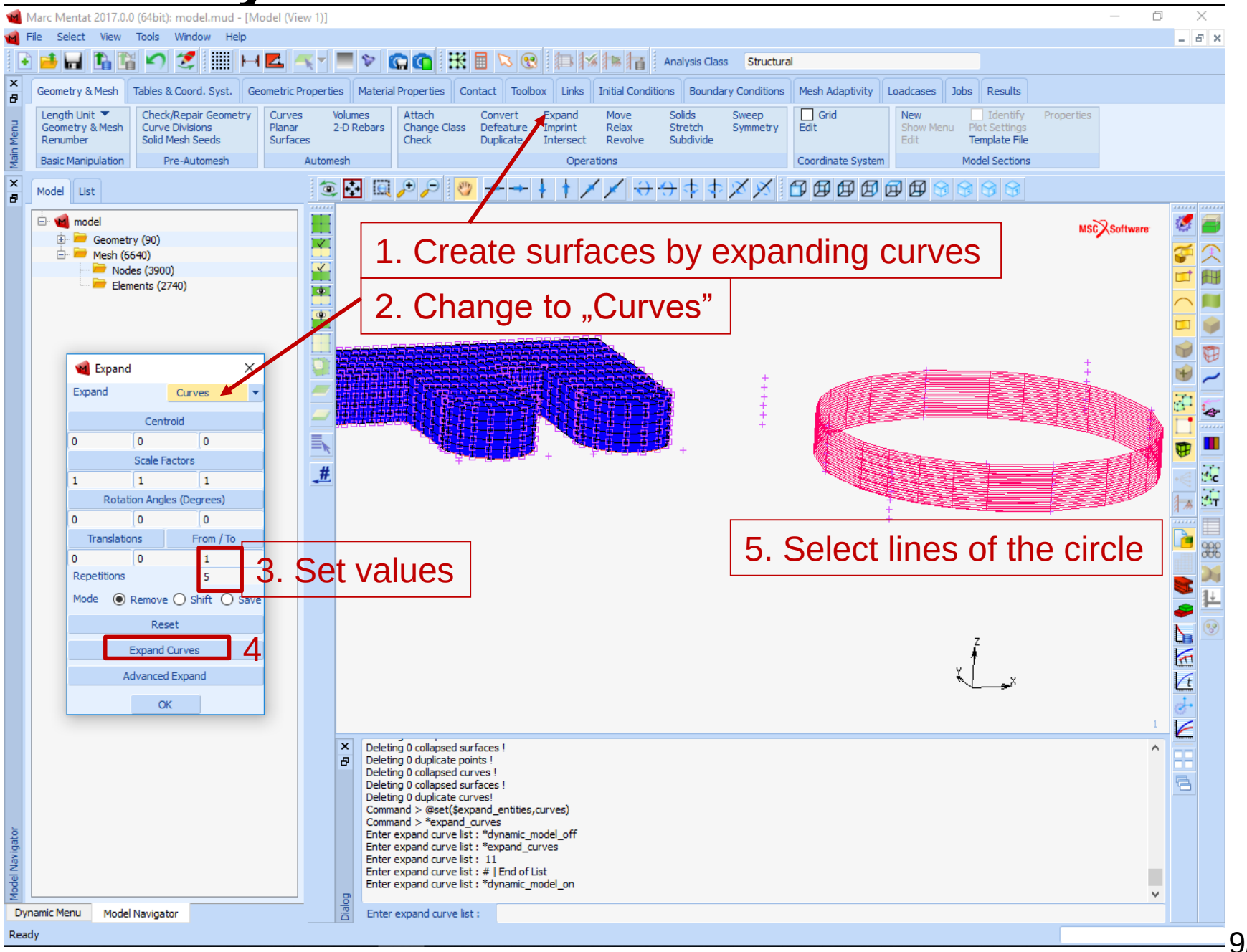

1. Add point: center of circle
2. Add circle using „Circle Cen/Rad” Curve
Geometry & Mesh
Circle Cen/Rad
2
1
Enter circle radius : 18
Curve 11 added.
Enter center point coordinates : *fill_view
Enter center point coordinates :
Ready

Geometry

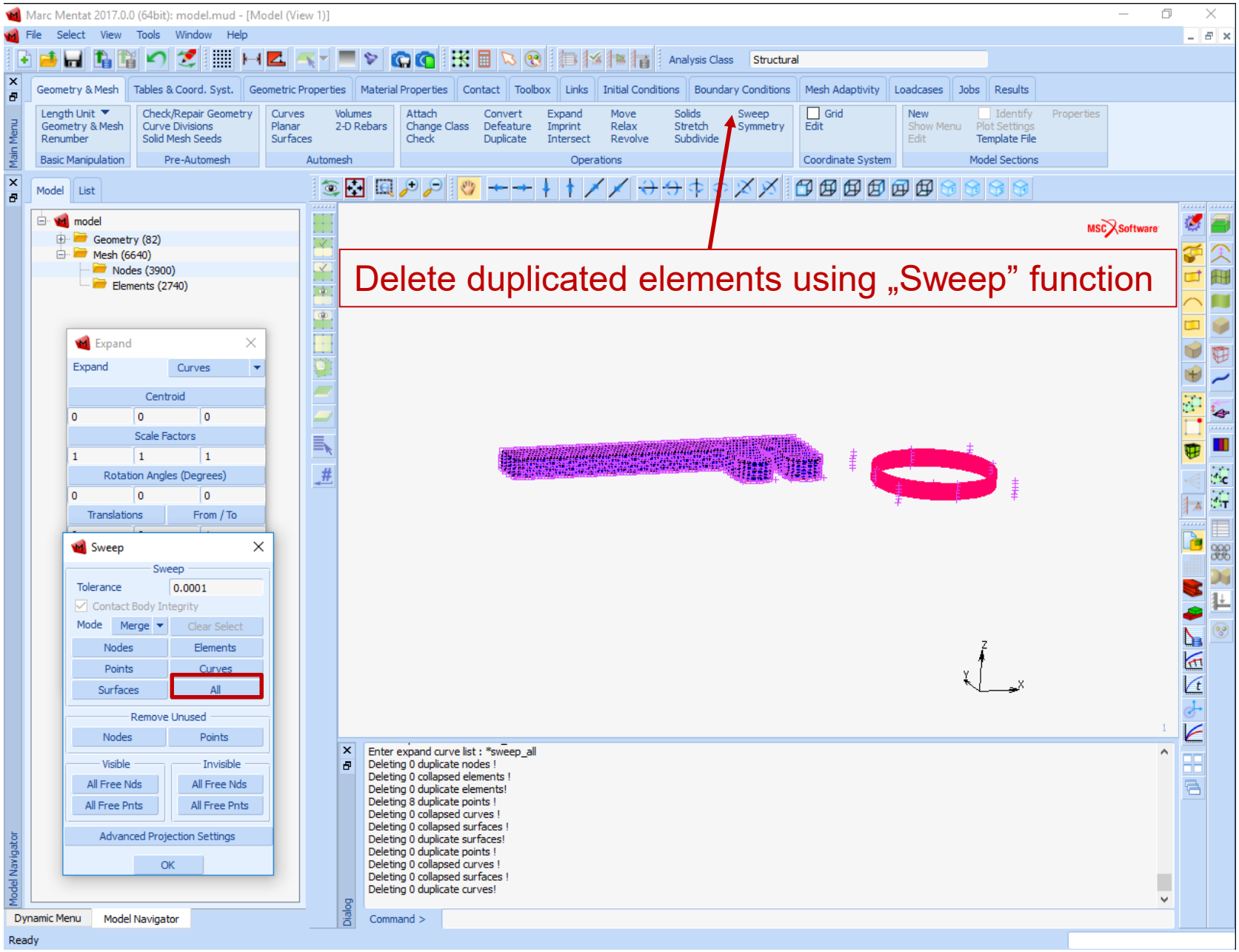


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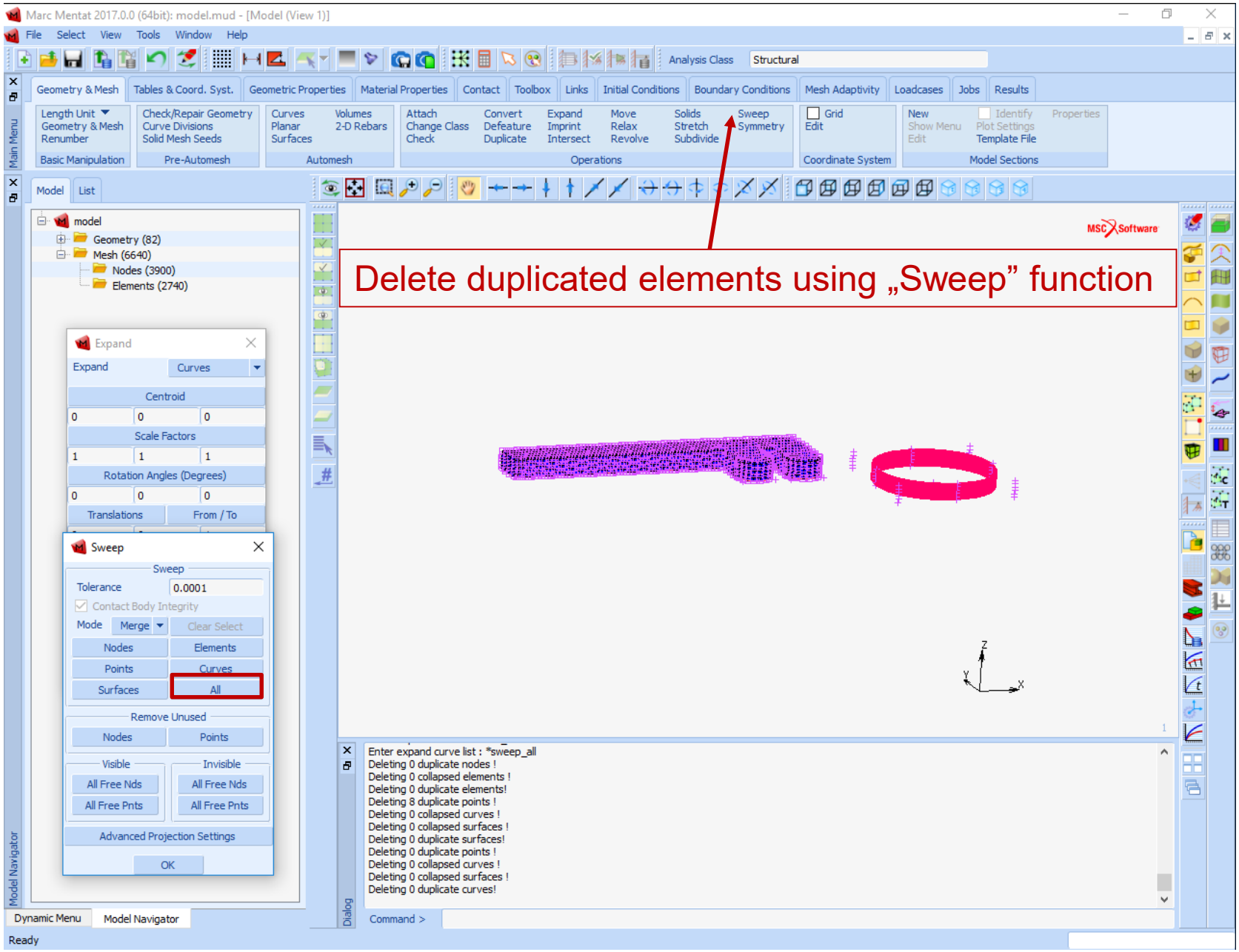
Geometry


1. Create surfaces by expanding curves
2. Change to „Curves”
3. Set values
4
5. Select lines of the circle
Expand
Curves
Centroid
Scale Factors
Rotation Angles (Degrees)
Translations
From / To
Repetitions
Mode
Remove
Shift
Save
Reset
Expand Curves
Advanced Expand
OK
Deleting 0 collapsed surfaces !
Deleting 0 duplicate points !
Deleting 0 collapsed curves !
Deleting 0 collapsed surfaces !
Deleting 0 duplicate curves!
Command > @set($expand_entities,curves)
Command > *expand_curves
Enter expand curve list : *dynamic_model_off
Enter expand curve list : *expand_curves
Enter expand curve list : 11
Enter expand curve list : # | End of List
Enter expand curve list : *dynamic_model_on
Enter expand curve list :
Ready

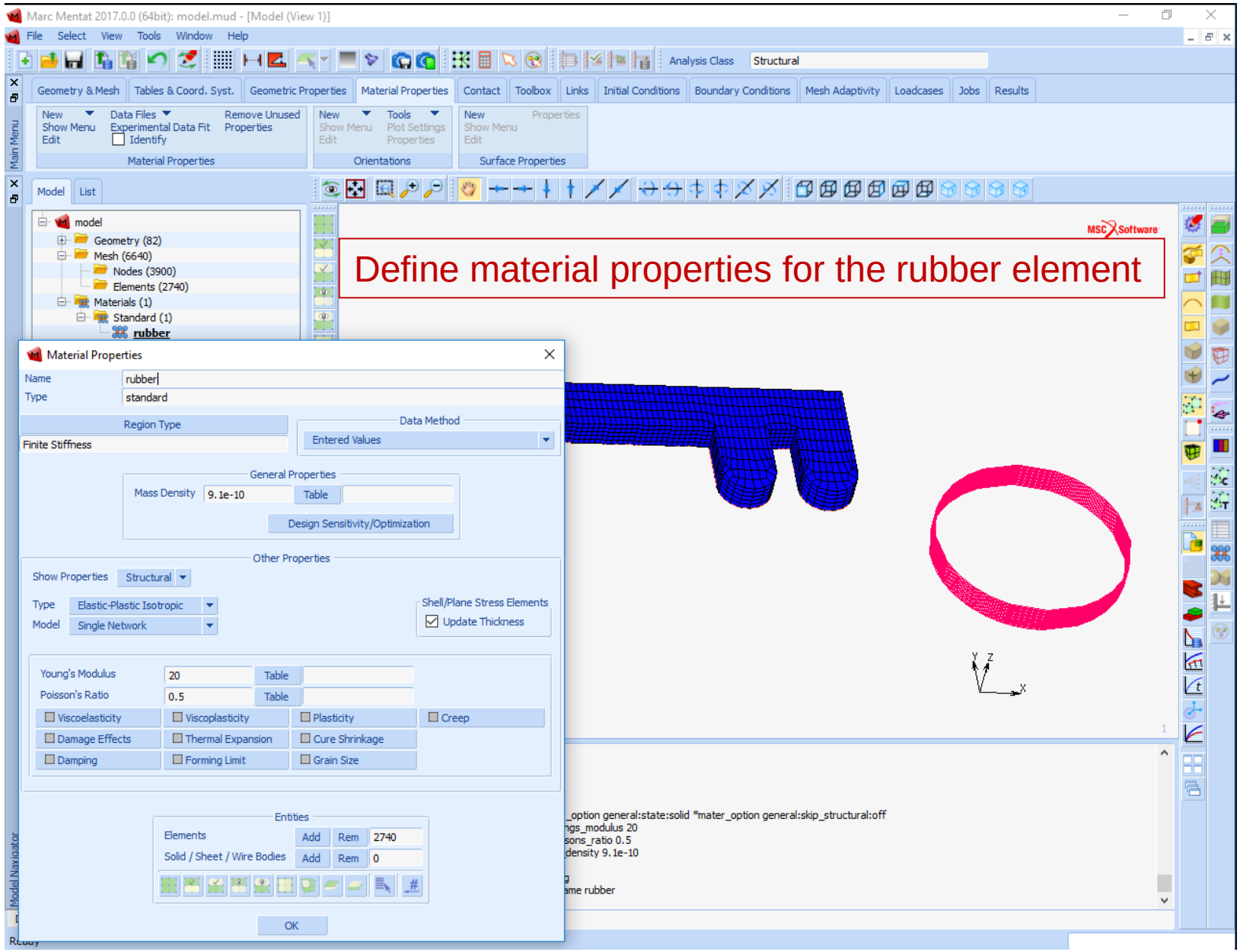
Geometry


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Check/Repair Geometry
Curve Divisions
Solid Mesh Seeds
Pre-Automesh
Curves
Planar
Surfaces
Volumes
2-D Rebars
Automesh
Attach
Change Class
Check
Convert
Defeature
Duplicate
Expand
Imprint
Intersect
Move
Relax
Revolve
Solids
Stretch
Subdivide
Sweep
Symmetry
Operations
Grid
Edit
Coordinate System
New
Show Menu
Edit
Identify
Plot Settings
Template File
Properties
Model Sections
Model
List
model
Geometry (82)
Mesh (6640)
Nodes (3900)
Elements (2740)
Delete duplicated elements using „Sweep” function
Expand
Expand Curves
Centroid
Scale Factors
Rotation Angles (Degrees)
Translations From / To
Sweep
Tolerance 0.0001
Contact Body Integrity
Mode Merge Clear Select
Nodes Elements
Points Curves
Surfaces All
Remove Unused
Nodes Points
Visible
Invisible
All Free Nds
All Free Pnts
Advanced Projection Settings
OK
Enter expand curve list : *sweep_all
Deleting 0 duplicate nodes !
Deleting 0 collapsed elements !
Deleting 0 duplicate elements!
Deleting 8 duplicate points !
Deleting 0 collapsed curves !
Deleting 0 collapsed surfaces !
Deleting 0 duplicate surfaces!
Deleting 0 duplicate points !
Deleting 0 collapsed curves !
Deleting 0 collapsed surfaces !
Deleting 0 duplicate curves!
Command >
Dynamic Menu
Model Navigator
Ready

Geometry

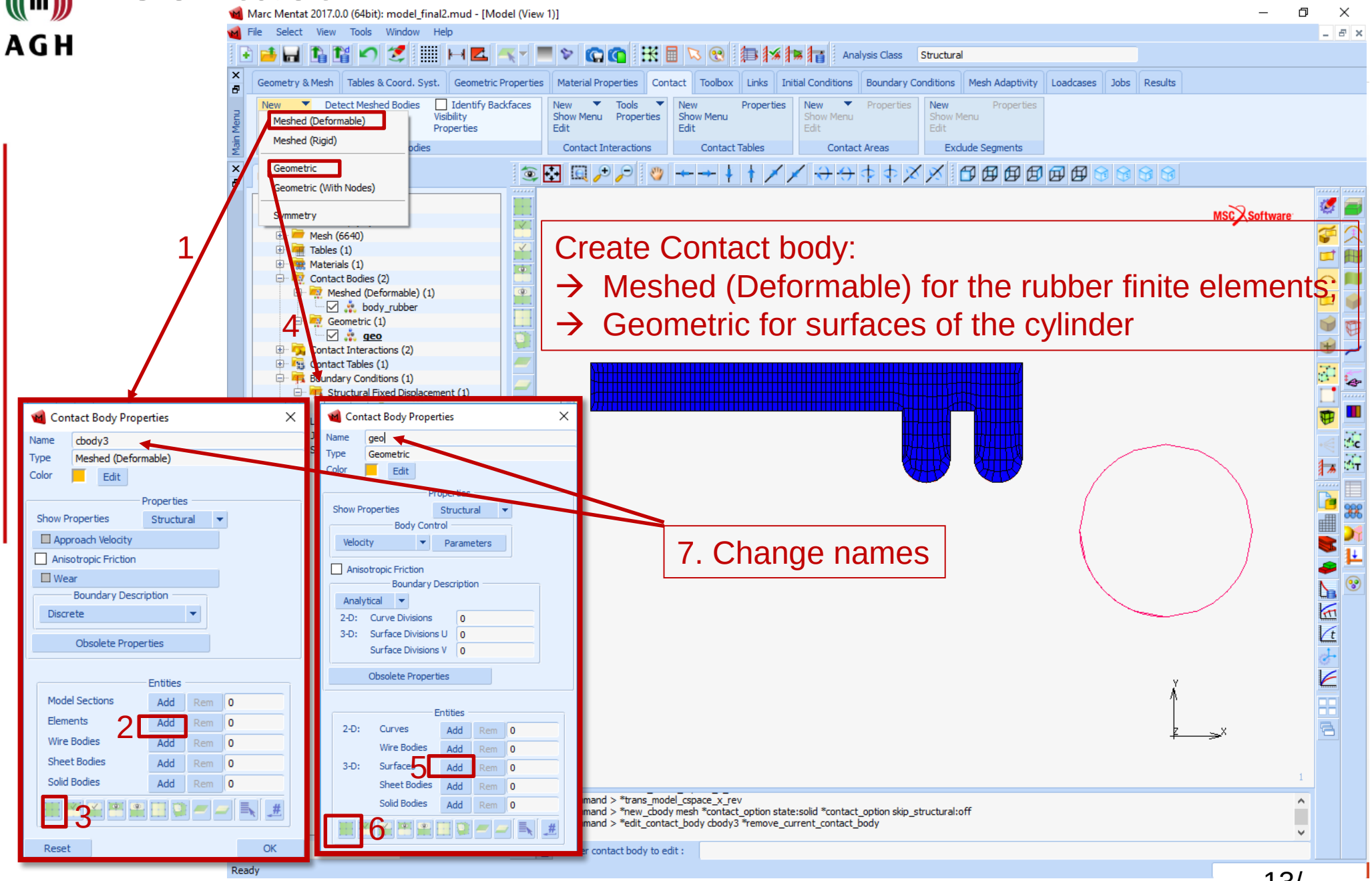

Delete duplicated elements using „Sweep” function
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Jobs
Results
Model
List
model
Geometry (82)
Mesh (6640)
Nodes (3900)
Elements (2740)
Expand
Curves
Centroid
Scale Factors
Rotation Angles (Degrees)
Translations
From / To
Sweep
Tolerance 0.0001
Contact Body Integrity
Mode Merge
Clear Select
Nodes
Elements
Points
Curves
Surfaces
All
Remove Unused
Visible
Invisible
All Free Nds
All Free Pnts
Advanced Projection Settings
OK
Enter expand curve list : *sweep_all
Deleting 0 duplicate nodes !
Deleting 0 collapsed elements !
Deleting 0 duplicate elements!
Deleting 8 duplicate points !
Deleting 0 collapsed curves !
Deleting 0 collapsed surfaces !
Deleting 0 duplicate surfaces!
Deleting 0 duplicate points !
Deleting 0 collapsed curves !
Deleting 0 collapsed surfaces !
Deleting 0 duplicate curves!
Command >
Dynamic Menu
Model Navigator
Ready

Material Properties



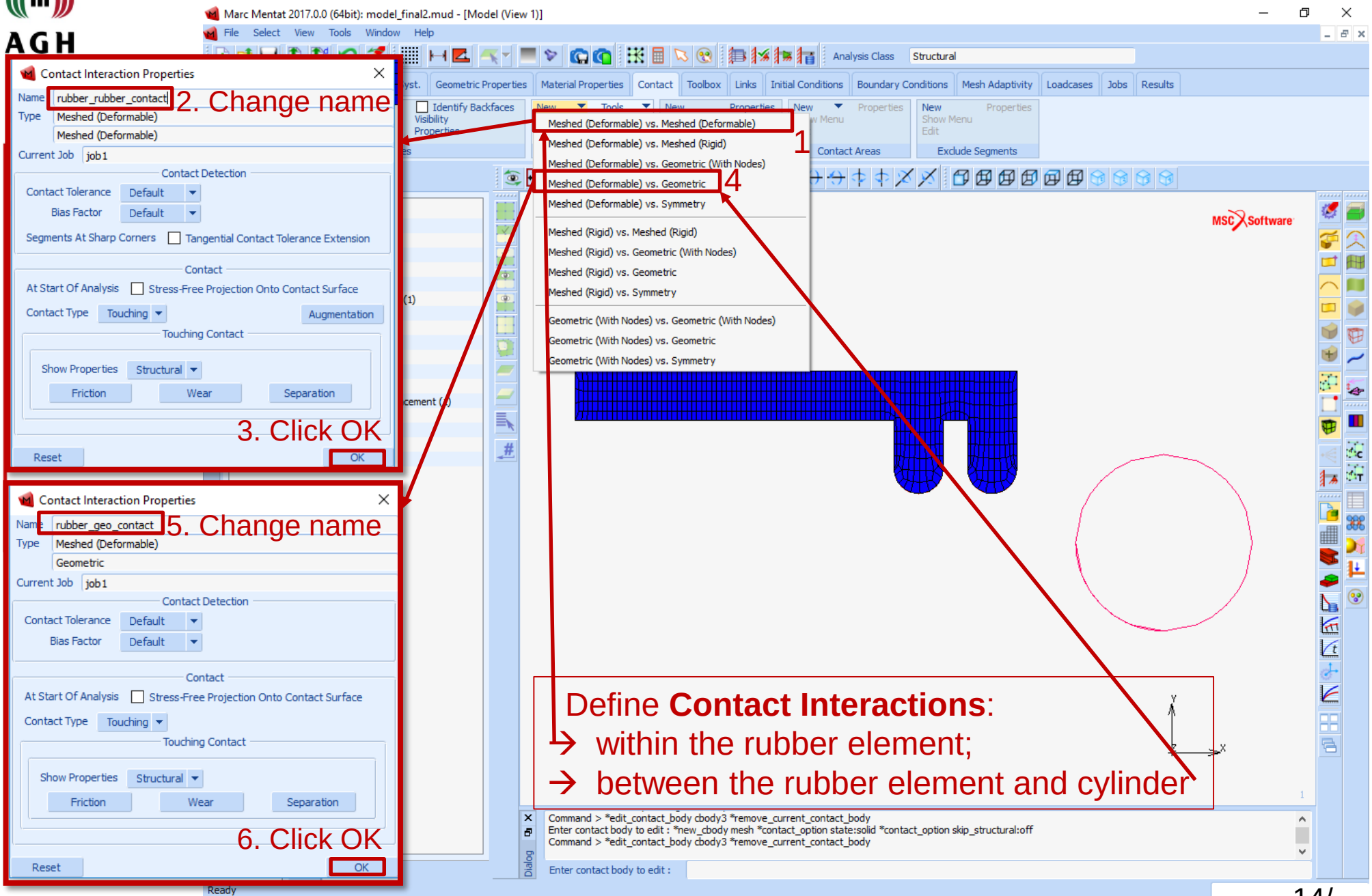
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Contact



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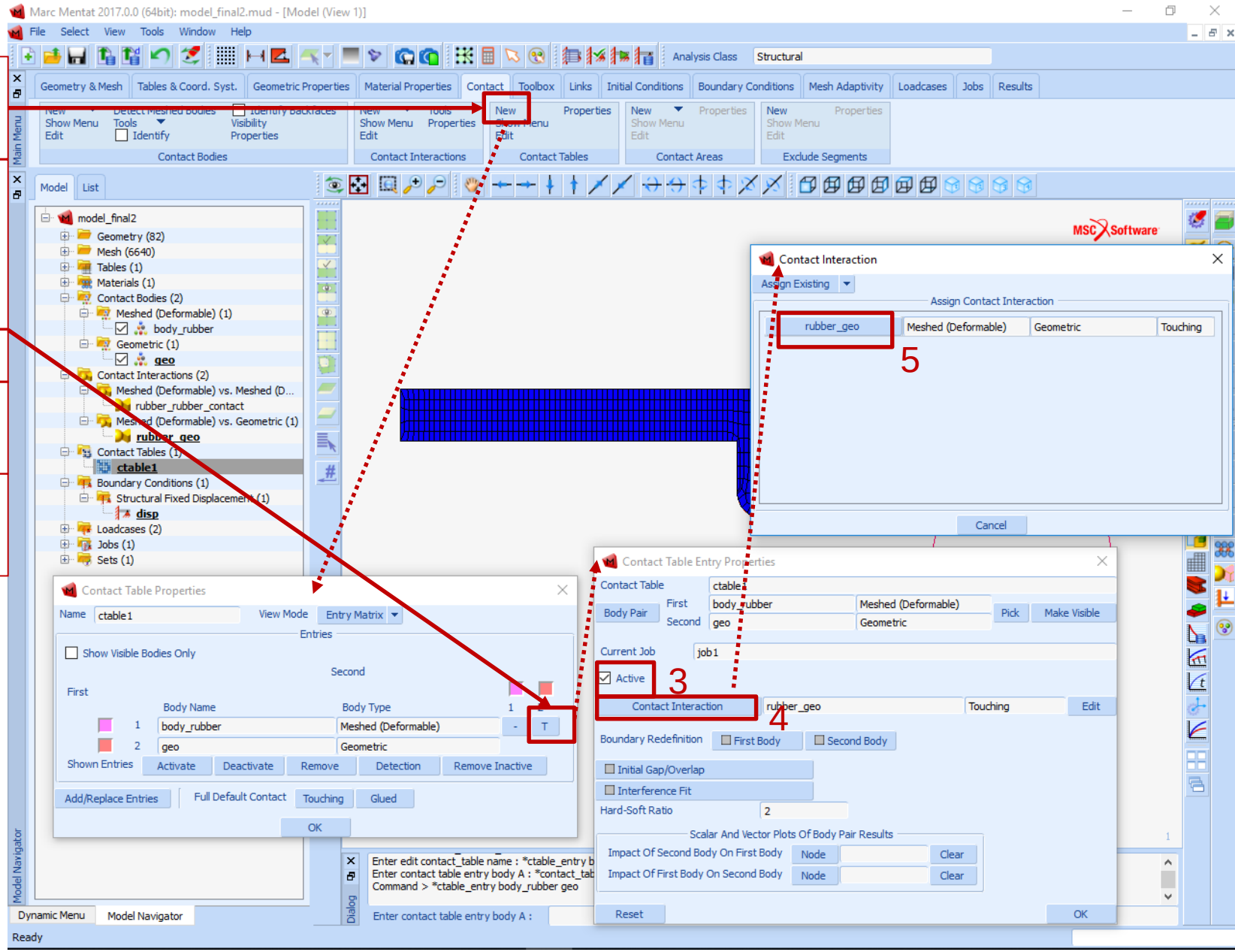
Contact



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Contact

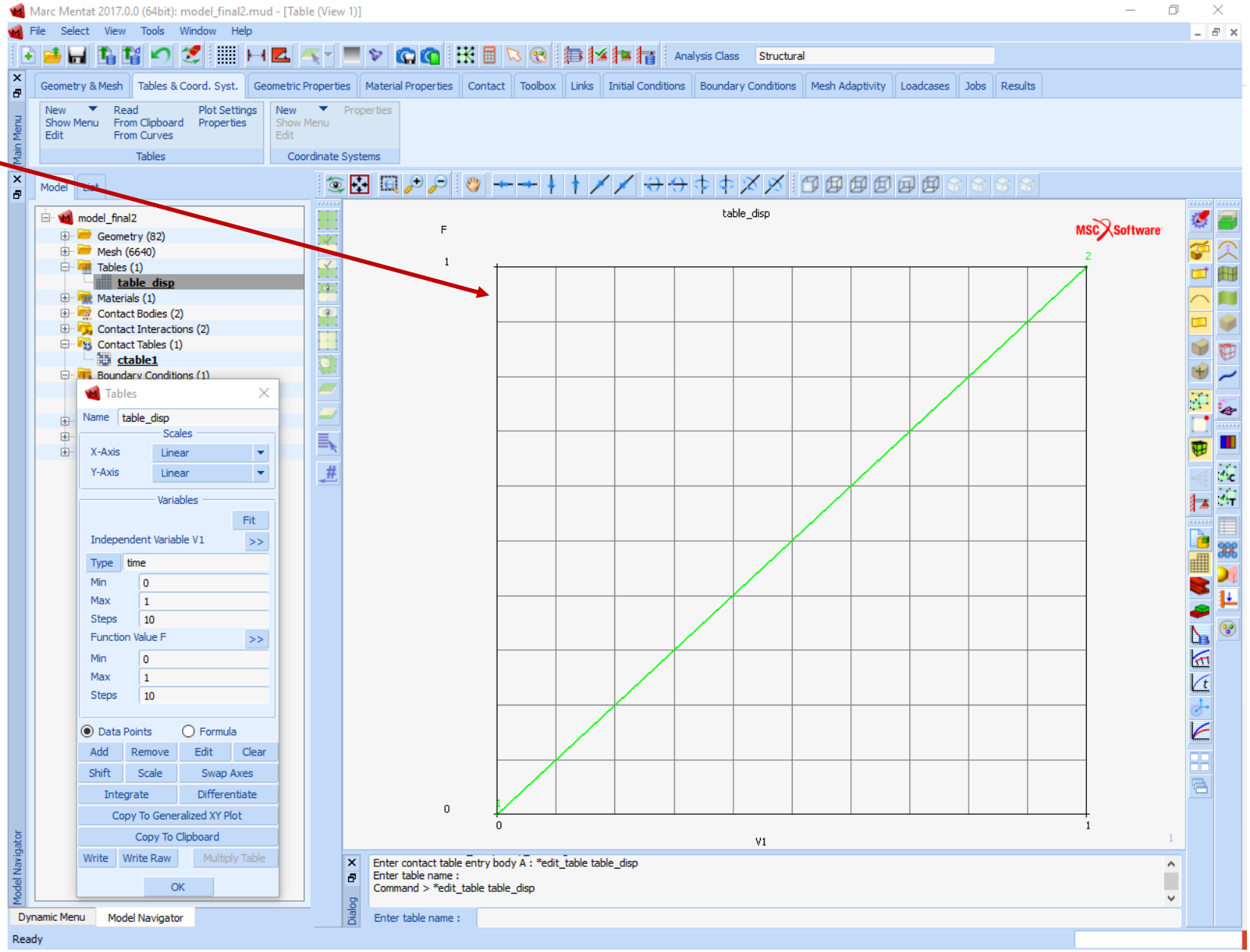
1. Create **Contact Table**
2. Create interaction between the rubber element and cylinder.
3. Tip „Active”.
4. Click „Contact Interaction” buton.
5. Choose available option.



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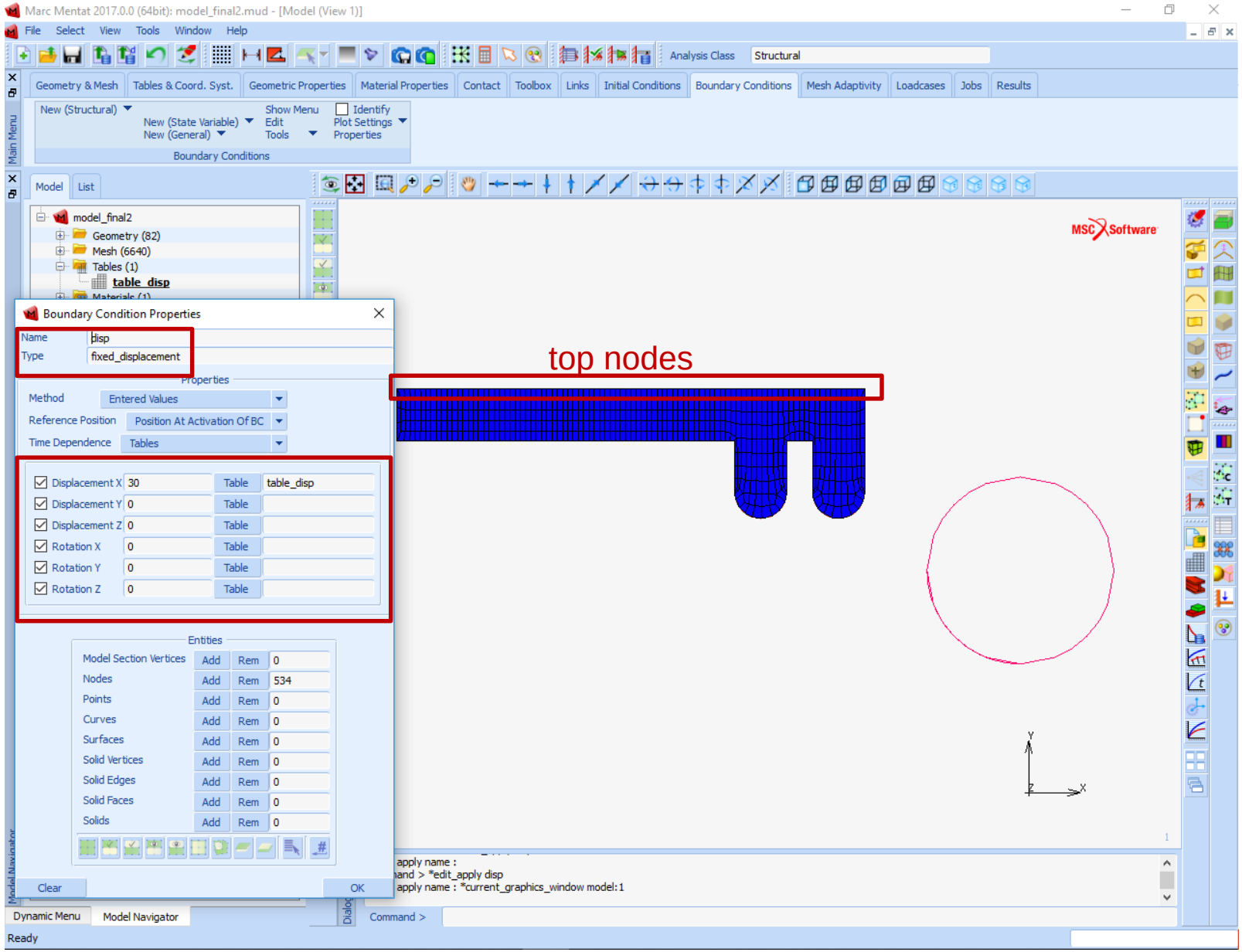
Boundary condition

Define table



Boundary condition

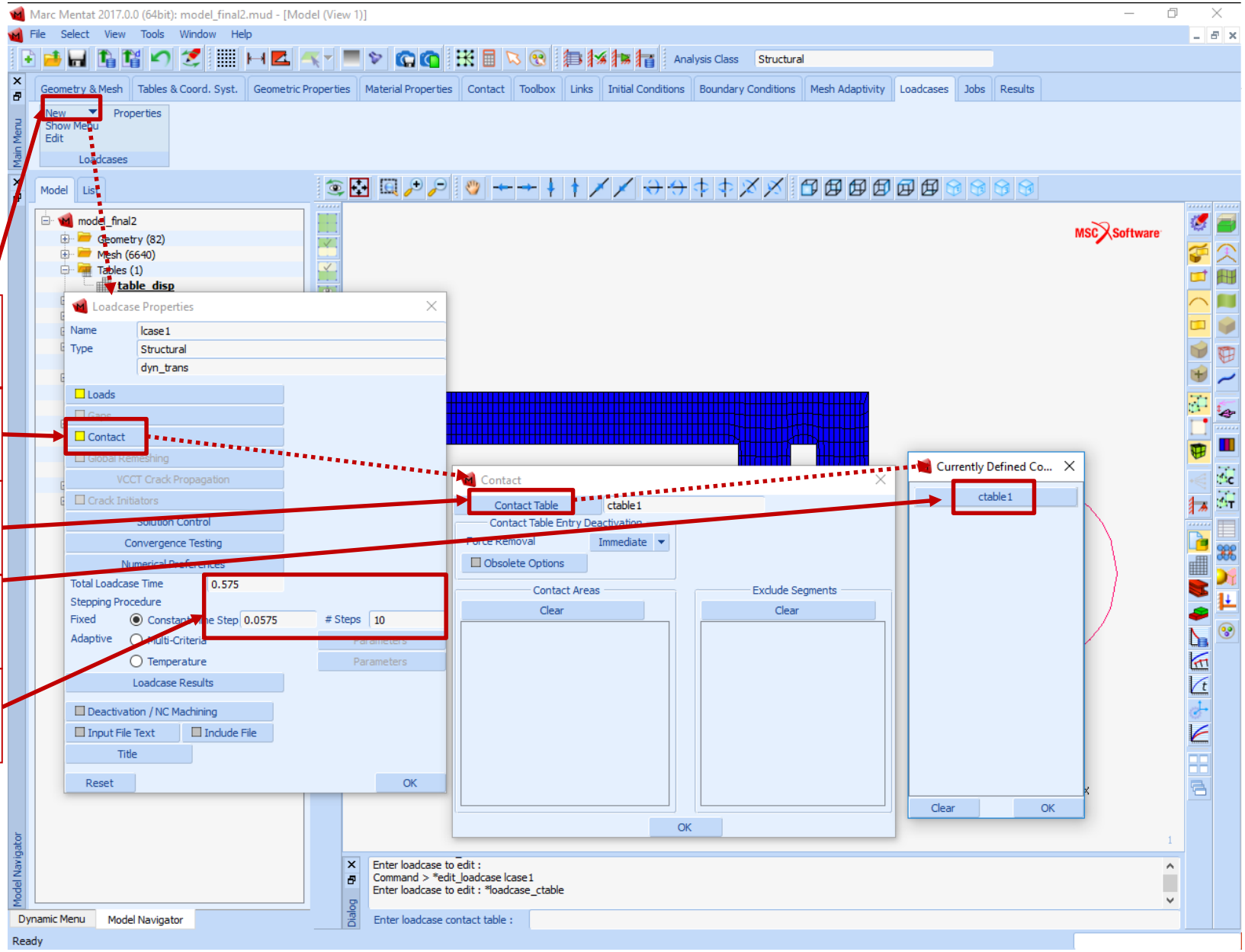
Define BC as showed for the top nodes of the rubber element.



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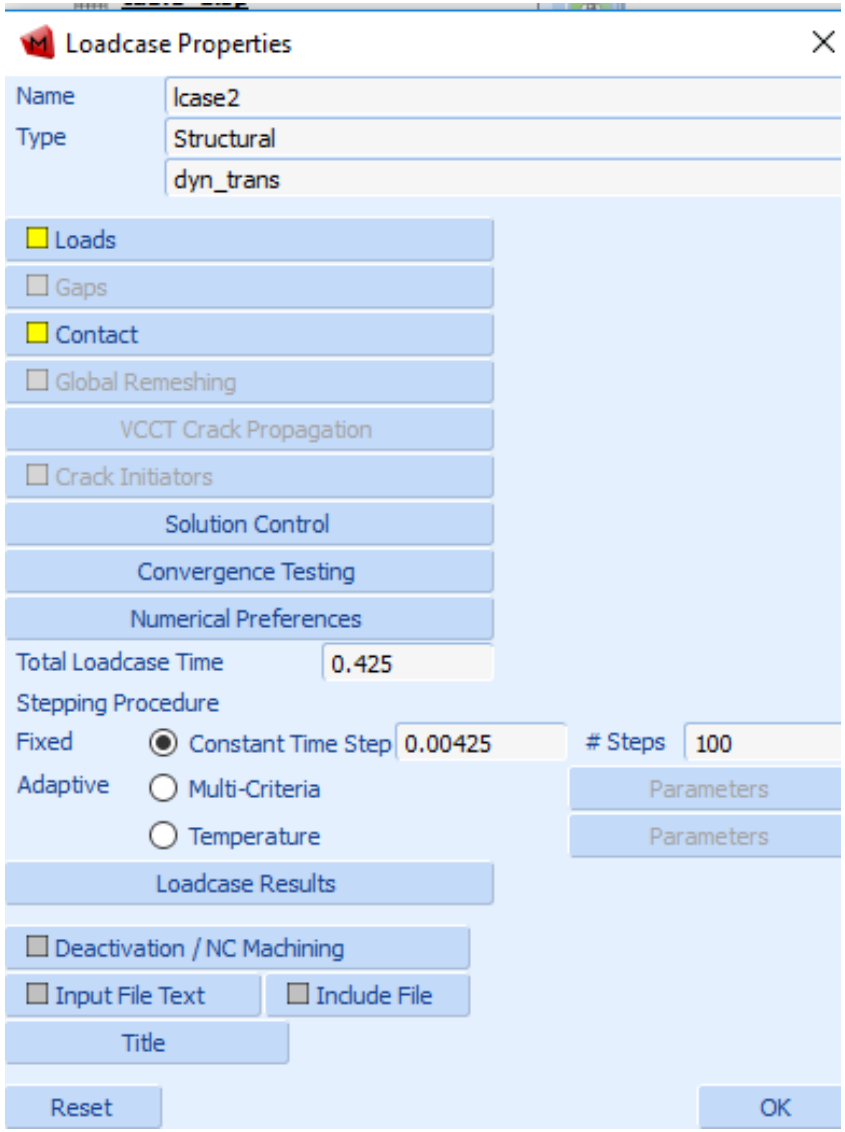
Loadcases

1. Create **Dynamic Transient** loadcase.
2. Click „Contact” button.
3. Click „Contact Table” button.
4. Click available option.
5. Set time and steps for loadcase.



Loadcases

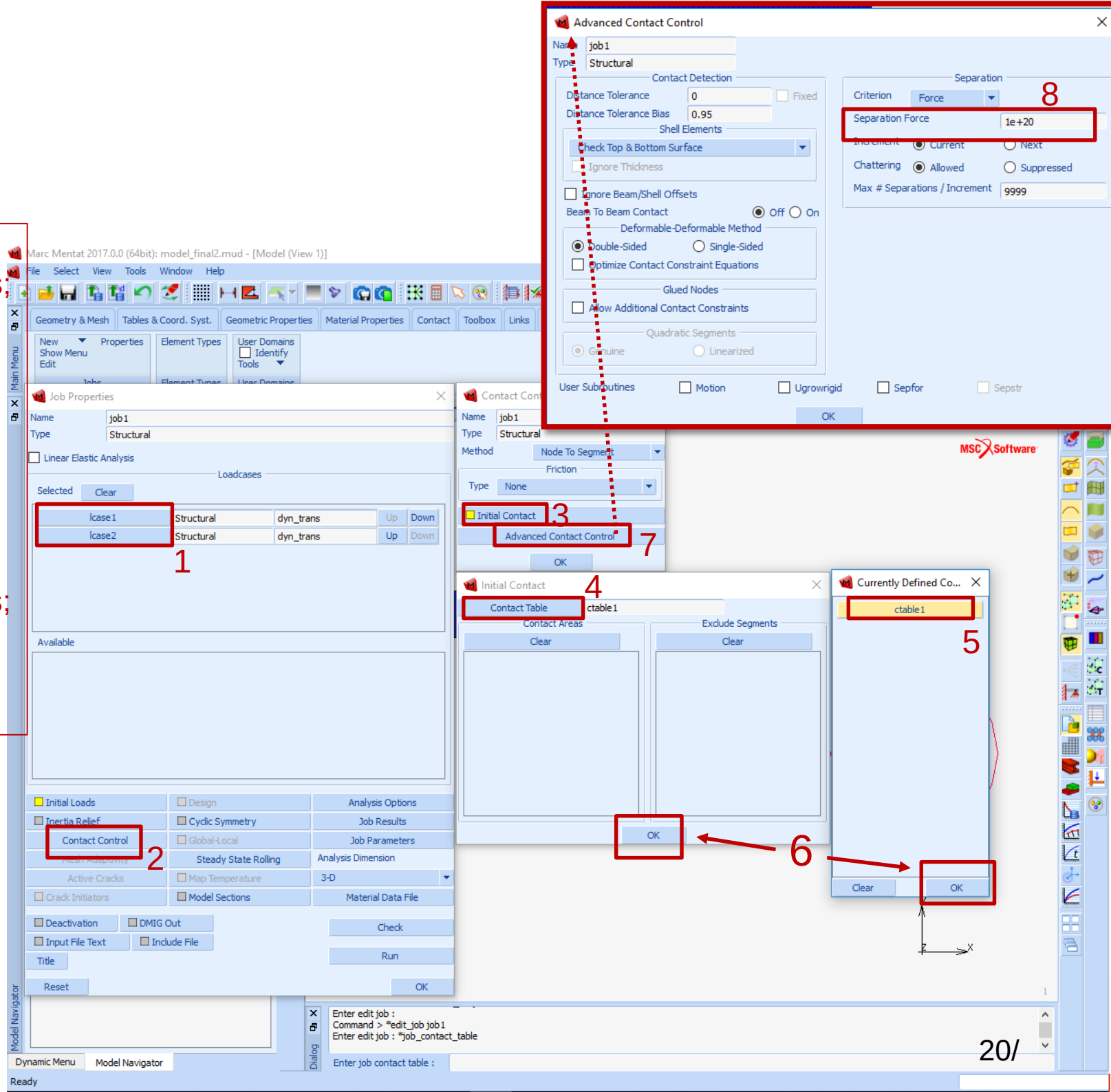
Repeat steps from previous slide for second loadcase.



Job

Define Job:

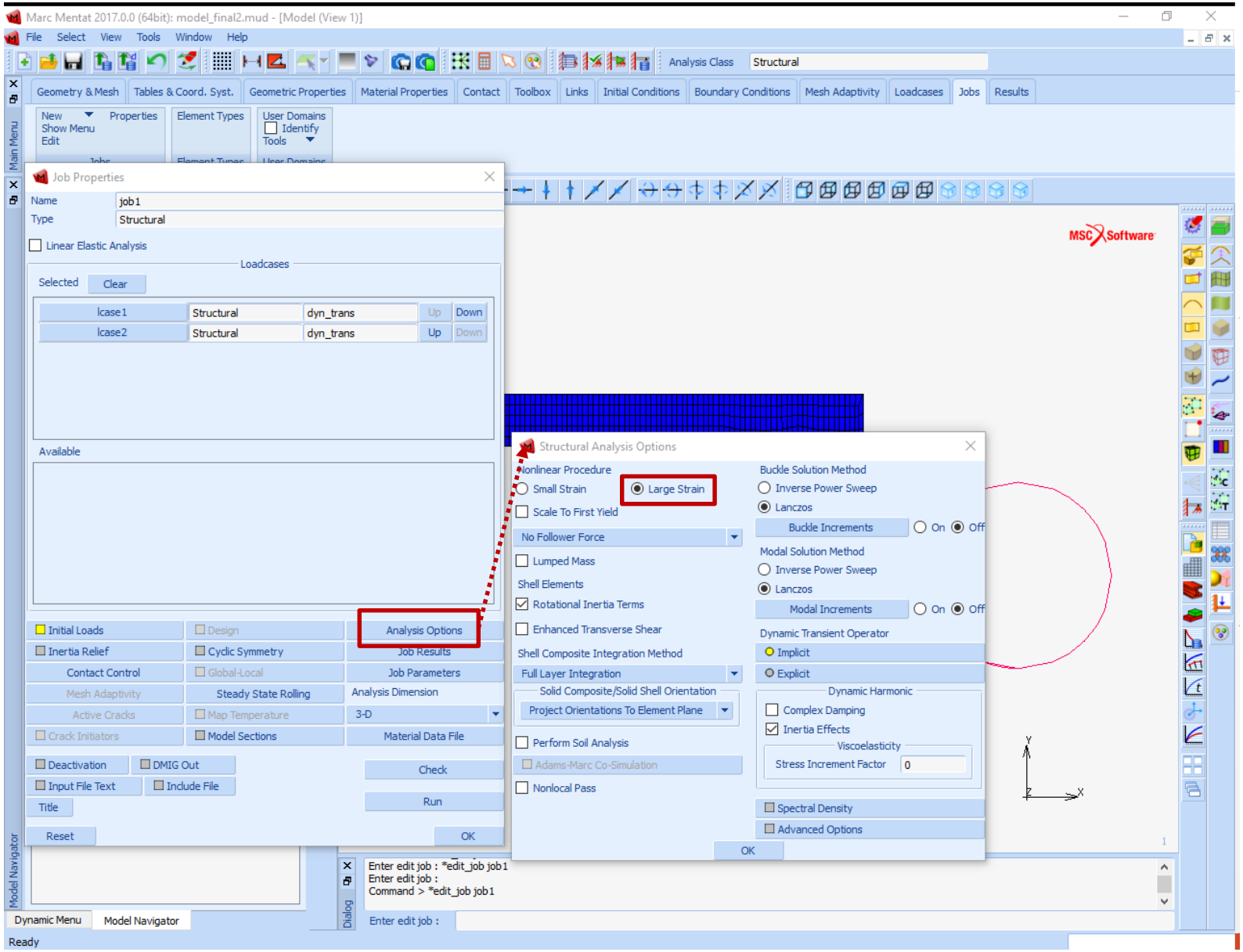
1. Add two loadcases;
2. Click „Contact Control” button;
3. Click „Initial Contact” button;
4. Click „Contact Table” button;
5. Choose available option;
6. Click OK two times;
7. Click „Advanced Contact Control”
8. Set value: 1e20



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Job

Set **Large Strain**



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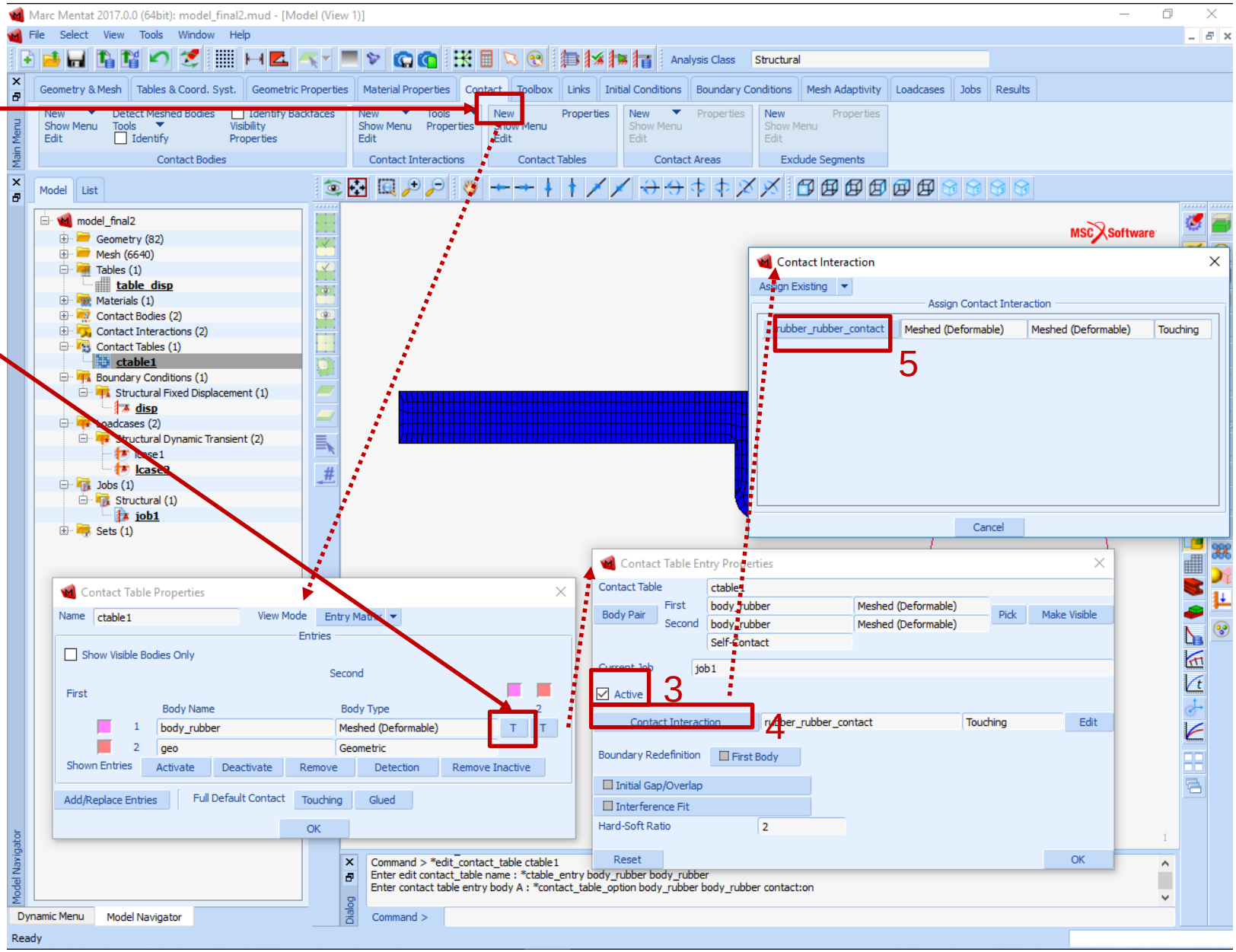
Job

RUN JOB AND ATTACH RESULTS TO REPORT

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Contact

1. Create **Contact Table**
2. Create interaction within the rubber element.
3. Tip „Active”.
4. Click „Contact Interaction” buton.
5. Choose available option.



Job

RUN JOB AND ATTACH RESULTS TO REPORT

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